Spinning top research

X25 (2025) — English translation

This problem is devoted to the study of the movements of symmetrical tops - rigid bodies with an axis of symmetry. In Part A, the general equations of motion of a rigid body will be obtained; in subsequent parts, the Euler top (which is not affected by external forces) and Lagrange top (moving in a constant gravitational field) will be considered. Let the bottom point of the top O, lying on the axis of symmetry, be fixed, and the top can freely rotate around the point O in all directions. The mass of the top is m, the distance from O to the center of mass of the top is l. To describe the motion, we introduce a fixed coordinate system $OXYZ$.

Part A. Euler's equations (2.2 points)

It is known from mechanics that the motion of an arbitrary rigid body with a fixed point is completely determined by a single vector - angular velocity, which means that the angular momentum and kinetic energy of a rigid body are uniquely expressed through its angular velocity and its geometric characteristics. Let us consider a small fragment of a solid body having a mass dm and a radius vector $\vec{r} = (X, Y, Z)$ relative to the point O in the coordinate system $OXYZ$. Let at the moment under consideration the body rotate with angular velocity $\vec{\Omega} = (\Omega_x, \Omega_y, \Omega_z)$.

A1 Write down in vector form the expression for the instantaneous speed $\vec{v}$ of the fragment dm at the moment under consideration. Express your answer in terms of $\vec{\Omega}$, $\vec{r}$. **0.20**

A2 Write down in vector form the expression for the angular momentum $d\vec{L}$ of the fragment dm at the moment under consideration. Express your answer in terms of $\vec{\Omega}$, $\vec{r}$, dm. **0.20**

A3 Obtain expressions for the components of the vector $d\vec{L} = (dL_x, dL_y, dL_z)$ in the coordinate system $OXYZ$. Express your answer in terms of X, Y, Z, Ω_x, Ω_y, Ω_z, dm. **0.30**

A4 For arbitrary motion of a rigid body, we can write angular momentum in the form **0.50**

$$
\begin{aligned}
L_x &= I_{xx}\Omega_x + I_{xy}\Omega_y + I_{xz}\Omega_z;\\
L_y &= I_{yx}\Omega_x + I_{yy}\Omega_y + I_{yz}\Omega_z;\\
L_z &= I_{zx}\Omega_x + I_{zy}\Omega_y + I_{zz}\Omega_z.
\end{aligned}
$$

Obtain expressions for the coefficients I_{xx}, I_{xy}, I_{xz}, I_{yx}, I_{yy}, I_{yz}, I_{zx}, I_{zy}, I_{zz} in the form of integrals over dm. The integral over dm is the sum over infinitesimal regions of a solid body of mass dm. For example, the coordinates of the center of mass can be written as

$$X_c = \frac{1}{m}\int X dm; \quad Y_c = \frac{1}{m}\int Y dm, \quad Z_c = \frac{1}{m}\int Z dm.$$

In further paragraphs, the formulas obtained here are not used.

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В дальнейших пунктах полученные здесь формулы не используются.

Можно показать, что кинетическая энергия твердого тела имеет вид

$$T = \frac{1}{2}\vec{L} \cdot \vec{\Omega}.$$

Используем следующий математический факт. Оказывается, для любого твёрдого тела поворотом осей можно добиться того, что $I_{xy} = I_{yz} = I_{zx} = 0$. В этом случае выражения для $\vec{L}$ и T сильно упрощаются, а I_{xx}, I_{yy}, I_{zz} называются главными моментами инерции . Для краткости обозначим $[\![M10]\!]$ Однако в этом случае возникает другая сложность: оси оказываются связаны с самим телом, и поэтому являются подвижными. Направим ось Oz вдоль оси симметрии волчка. Выберем оси Ox, Oy так, чтобы они вместе с осью Oz образовывали правую тройку. Из соображений симметрии очевидно, что в этом случае оси $Oxyz$ являются главными для волчка.

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$$I_{xx} = I_x, \qquad I_{yy} = I_y, \qquad I_{zz} = I_z.$$

Однако в этом случае возникает другая сложность: оси оказываются связаны с самим телом, и поэтому являются подвижными.

Направим ось Oz вдоль оси симметрии волчка. Выберем оси Ox, Oy так, чтобы они вместе с осью Oz образовывали правую тройку. Из соображений симметрии очевидно, что в этом случае оси $Oxyz$ являются главными для волчка.

Пусть $\vec{e}_x$, $\vec{e}_y$, $\vec{e}_z$ – единичные векторы осей $Oxyz$. Пусть в некоторый момент волчок вращается с угловой скоростью $\vec{\omega}$, причём в системе координат $Oxyz$, связанной с волчком, $\vec{\omega} = \omega_x\vec{e}_x + \omega_y\vec{e}_y + \omega_z\vec{e}_z$.

A5	Derive expressions for $\vec{L}$ and T in the principal axes. Express your answer in terms of I_x, I_y, I_z, ω_x, ω_y, ω_z, $\vec{e}_x$, $\vec{e}_y$, $\vec{e}_z$.	**0.50**

To study rotations, the basic equation of rotational motion is used:

$$\dot{\vec{L}} = \vec{M},$$

где $\vec{L}$ – момент импульса системы относительно выбранной неподвижной точки, $\vec{M}$ – суммарный момент внешних сил относительно этой точки. Поскольку оси системы координат $Oxyz$ вращаются с угловой скоростью $\vec{\omega}$, производные единичных векторов подвижной системы координат имеют вид

$$\dot{\vec{e}}_x = \vec{\omega} \times \vec{e}_x, \quad \dot{\vec{e}}_y = \vec{\omega} \times \vec{e}_y, \quad \dot{\vec{e}}_z = \vec{\omega} \times \vec{e}_z.$$

A6	Obtain expressions for the components of the vector $\dot{\vec{L}}$ in the moving coordinate system $\left(\dot{\vec{L}}\right)_x$, $\left(\dot{\vec{L}}\right)_y$, $\left(\dot{\vec{L}}\right)_z$. Express your answer in terms of I_x, I_y, I_z, ω_x, ω_y, ω_z, $\dot{\omega}_x$, $\dot{\omega}_y$, $\dot{\omega}_z$.	**0.50**

From point A6 it follows that the basic equation of dynamics in the moving axes $Oxyz$ has the form

$$\begin{cases} I_x\dot{\omega}_x + (I_z - I_y)\omega_y\omega_z = M_x \\ I_y\dot{\omega}_y + (I_x - I_z)\omega_z\omega_x = M_y \\ I_z\dot{\omega}_z + (I_y - I_x)\omega_x\omega_y = M_z \end{cases}$$

Здесь M_x, M_y, M_z – компоненты внешнего момента $\vec{M}$ в системе координат $Oxyz$. These equations are called dynamic Euler equations and are a very convenient tool for studying the rotation of a rigid body.

Part B. Equations of dynamics of a symmetrical top (2.1 points)

In the future, we will be interested in the movement of the center of mass of the top. The position of the top in space can be specified by two angles, similar to the angles in the spherical coordinate system. The angle between the axis OZ and the axis of symmetry of the top Oz will be denoted θ, the angle of rotation of the axis of symmetry of the top around OZ – φ. In addition, the top can rotate around its own axis of symmetry, which does not affect the position of its center of mass. Thus, the overall motion of the top is a superposition of three rotations: Rotation about the axis of symmetry with a certain angular velocity $\vec{\omega}_s$. The rotation at which the angle θ changes, the corresponding angular velocity $\vec{\omega}_\theta$ is perpendicular to the axes OZ and Oz. Rotation at which the angle φ changes, the corresponding angular velocity $\vec{\omega}_\varphi$ is directed along OZ. Then the total angular velocity of the top is the sum of these three angular velocities

$$\vec{\omega} = \vec{\omega}_s + \vec{\omega}_\theta + \vec{\omega}_\varphi .$$

Заметим, что в силу симметрии волчка мы можем выбрать оси Ox и Oy в плоскости, перпендикулярной оси симметрии волчка произвольным образом. Воспользуемся этим и направим ось Ox в заданный момент времени в плоскости OZz, так, чтобы угол $xOZ[\![M19]\!]$ \vec{\omega} = \vec{\omega}_s + \vec{\omega}_\theta + \vec{\omega}_\varphi .$$

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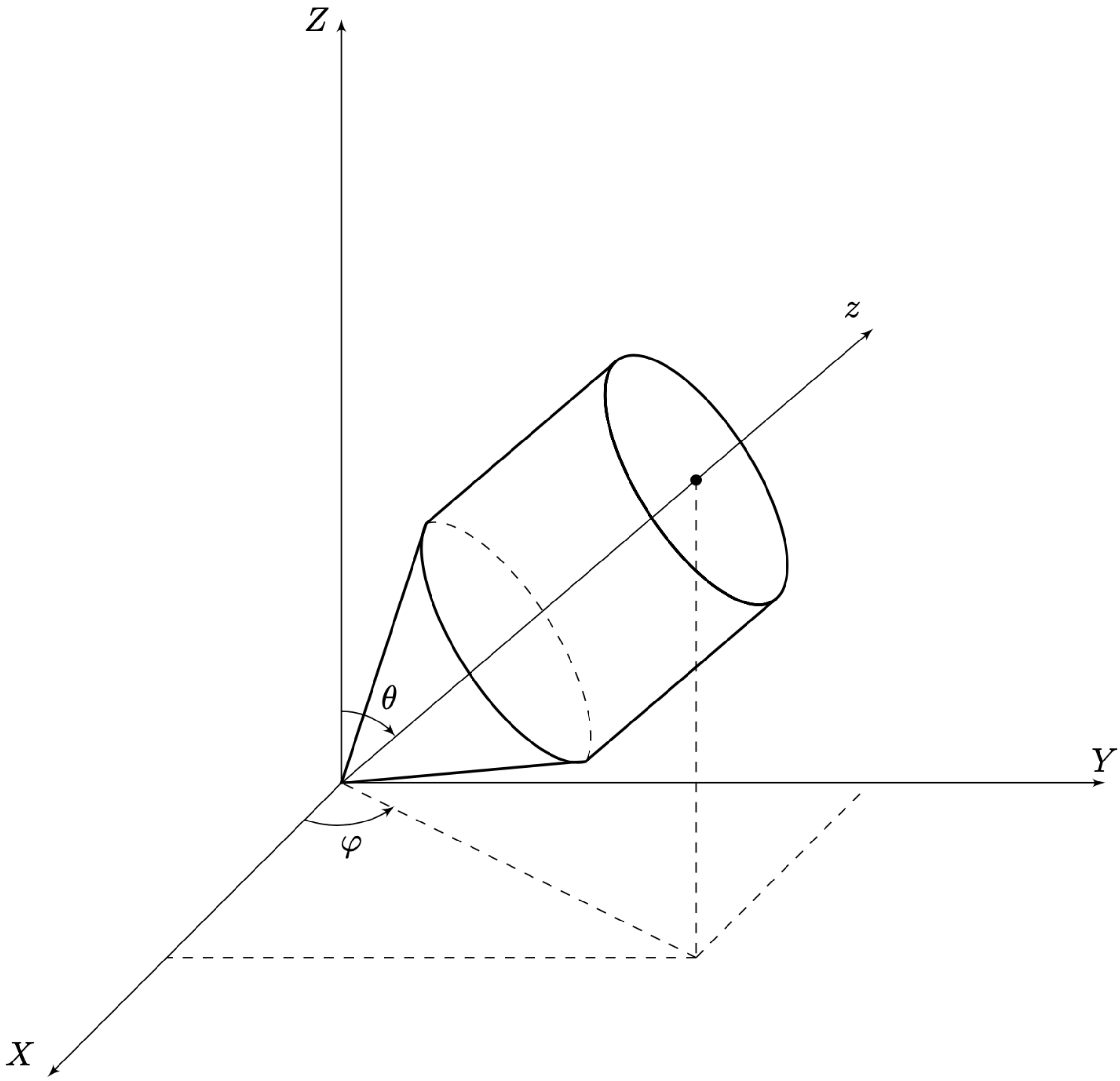

Fig. 1.

B1	Find the projections of angular velocity ω_x, ω_y, ω_z. Express your answer in terms of ω_s, $\dot{\theta}$, $\dot{\varphi}$, θ.	**0.40**

Let us consider the equations of the dynamics of a top in the case when it moves under the influence of gravity. Let the free fall acceleration be equal to g and directed against the axis OZ. Due to symmetry for the top under consideration $I_x = I_y$. We denote $I_x = I_y = A$, $I_z = C$ and will use these notations throughout.

B2	Show that the projection L_1 of the angular momentum $\vec{L}$ onto the fixed vertical axis OZ is constant. Show that the projection L_2 of the angular momentum $\vec{L}$ onto the moving axis Oz is constant.	**0.40**

If you were unable to prove the constancy of L_1, L_2, you can then use this fact without proof.

B3	Using the results of step A5, express the values of the projections L_1 and L_2 in terms of ω_s, $\dot{\theta}$, $\dot{\varphi}$, θ, A, C.	**0.60**

B4 At this point we will assume that $l = 0$, and the moment of gravity can be ignored. Let the angular momentum of the top be equal to L_0 and directed along the axis OZ. At time $t = 0$ the values of angles $\theta = \theta_0$, $\varphi = \varphi_0$. Find the dependence of angles θ, φ on time. This movement is called regular precession. **0.70**

Part C. Effective Lagrange top potential (1.5 points)

In this part we will study the movement of the top taking into account the moment of gravity.

C1 For an arbitrary position of the top, express the total energy E of the motion of the top (sum of kinetic and potential) in terms of ω_s, $\dot{\theta}$, $\dot{\varphi}$, A, C, m, g, l, θ. **0.60**

C2 Express ω_s and $\dot{\varphi}$ in terms of L_1, L_2, A, C, θ using the results from step B3. **0.30**

C3 Using the constancy of L_1, L_2, reduce the expression for the total energy E to the form **0.60**

$$E = \frac{\mu}{2}\dot{\theta}^2 + U(\theta).$$

Получите выражения для μ, $U(\theta)$ через A, C, L_1, L_2, m,g, l. Note. As is known, the total energy is determined up to a constant, which for convenience can be set equal to zero.

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From the above it follows that the study of the time dependence of θ reduces to the study of a one-dimensional problem with energy E, and $\frac{\mu}{2}\dot{\theta}^2$ can be considered as kinetic energy, and $U(\theta)$ as potential. Therefore, in this case, $U(\theta)$ is called the effective potential.

Part D. Various movements of the Lagrange top (6.2 points)

In this part we will look at several main possible modes of motion of a Lagrange top.

Physical pendulum Let at the initial moment $\theta = \theta_0 > 0$, and the top is motionless. The top is released without initial speed.

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D1 Find the period of oscillation of the top. You will receive the answer in the form of a definite integral containing A, C, l, θ_0, g. Also obtain a simplified expression for the case $\pi - \theta_0 \ll 1$. **0.80**

“Sleeping” top Another obvious movement of the top is the situation when $\theta = 0$ is during the entire movement. Such a top is called “sleeping” due to the lack of movement of its center of mass. Let the “sleeping” top rotate with angular velocity ω_0 around its axis.

Another obvious movement of the top is the situation when $\theta = 0$ is in the process of the entire movement. Such a top is called “sleeping” due to the lack of movement of its center of mass.

Let the “sleeping” top rotate with angular velocity ω_0 around its axis.

D2 Express L_1 and L_2 in terms of A, C, ω_0. **0.20**

Let now the "sleeping" top be slightly deflected from the vertical position, while the rotation parameters are selected so that L_1 and L_2 remain equal to those found in the previous paragraph.

D3 Find a condition relating A, C, m, g, l, ω_0 such that the position $\theta = 0$ is stable for the deviations described above. This condition is called Maievsky's condition. It turns out that if the found condition is satisfied, the equilibrium position is stable even if the conservation condition L_1 and L_2 is abandoned. **0.50**

Regular precession It can be shown that for the potential $U(\theta)$ for any L_1, L_2 there is a single angle θ_1, which is the equilibrium position for the equivalent one-dimensional system.

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D4 Obtain an equation for θ that is satisfied by the angle θ_1. The equation may contain A, C, L_1 , L_2, m, g, l. Clue. $U(\theta)$ can be represented as a function of the new variable $s = \cos\theta$, which greatly simplifies the solution. **0.70**

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From point C2 it follows that at $\theta = \theta_1 = const$ the motion is a regular precession: $\dot{\varphi} = \omega_p = const$ is the angular velocity of precession, $\omega_s = const$ is the angular velocity of its own rotation.

D5 Find the possible precession rates of the top ω_p at $\theta = \theta_1$. Express your answer in terms of A , C, m, g, l, ω_s, θ_1. One of the solutions can be to substitute expressions for L_1, L_2 into the equation from D4 through the angular velocities ω_s, ω_p. **0.80**

D6 Obtain approximate expressions for the frequencies in the case of $mgl \ll C\omega_s^2$. Consider that A and C are quantities of the same order. **0.30**

Nutation and irregular precession In general, the angle θ does not remain constant. It can be shown that for the potential $U(\theta)$ for any L_1, L_2 during the motion θ periodically changes from some θ_{min} to some θ_{max}. This change in angle θ is called nutation. In this case, the angle φ may change unevenly, so the precession is called irregular. Let's consider the following situation: at the initial moment of time, the axis of the top is horizontal ($\theta = \pi/2$), the center of mass of the top is motionless, the top is twisted around the symmetry axis with an angular velocity ω_0. The top is released without a push.

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Let's consider the following situation: at the initial moment of time, the axis of the top is horizontal ($\theta = \pi/2$), the center of mass of the top is motionless, the top is twisted around the symmetry axis with an angular velocity ω_0. The top is released without a push.

D7 Determine the angle α_{max} of the maximum deviation of the top axis from the horizontal during the subsequent movement. Express your answer in terms of A, C, m, g, l, ω_0. **0.70**

Now let the top be quickly spun, i.e. $C\omega_0^2 \gg mgl$. Consider that A and C are quantities of the same order.

D8	Find the dependence of the angle of deviation of the top axis from the horizontal on time $\alpha(t)$. Express your answer in terms of A, C, m, g, l, ω_0.	**0.70**

D9	Find the dependence of the precession angle on time $\varphi(t)$. At the initial moment $\varphi(0) = \varphi_0$. Express your answer in terms of A, C, m, g, l, ω_0, φ_0.	**0.50**

D10	Draw a characteristic graph of the dependence $\alpha(\varphi)$. Determine the angle $\Delta\varphi$ by which the top axis rotates between two successive returns of the top axis to a horizontal position. Express your answer in terms of A, C, m, g, l, ω_0.	**0.50**

Let the top be a thin rod of mass m, length R, to the end of which a thin homogeneous disk of mass $2m$ and radius R is attached. The attachment point coincides with the center of the disk, the rod is perpendicular to the plane of the disk. As before, the top is released from a horizontal position without a push, spinning around its axis to an angular velocity $\omega_0 = \sqrt{100\frac{g}{R}}$.

D11	Calculate $\Delta\varphi$ for the described system.	**0.50**

Source: https://pho.rs/p/4199